

Motorized Safe Latch Control Circuit Package

Concept package for a keypad-controlled 12 V geared motor that locks or unlocks a latch with automatic end-of-travel stopping.

Goal

Design a conceptual 12 V motorized safe latch controller that locks or unlocks by reversing a geared DC motor. The user asked for a keypad-based design, with automatic end-of-travel stopping for safety.

Assumptions

- This is a conceptual electronics package for a hobby/prototype safe latch, not a certified physical security product.
- A low-speed 12 V DC geared motor rotates a cam or lead-screw that moves the latch between LOCKED and UNLOCKED positions.
- Two limit switches are mechanically positioned so they close at the ends of travel and inform the controller when the latch is fully locked or fully unlocked.
- A 4x4 matrix keypad provides the unlock/lock command input through firmware.
- The motor current is kept within the TB6612FNG driver capability; if the real actuator stalls above about 1 A average, a higher-current H-bridge is needed.
- The 12 V input comes from an external DC adapter or internal 12 V system and is stepped down to 5 V for logic.

Basic circuit description

The circuit is divided into five blocks: input power, logic power conversion, keypad/controller, motor driver, and end-of-travel sensing. A 12 V input rail feeds the geared latch motor directly through the motor driver and also feeds a 5 V buck converter for logic power. The ATtiny1616 microcontroller scans a 4x4 keypad, validates a PIN in firmware, and commands the H-bridge motor driver to run forward for unlock or reverse for lock. Two normally-open limit switches are wired to microcontroller inputs with pull-ups so that each end position is sensed independently. Firmware stops the motor immediately when the relevant limit switch becomes active, and can also impose a timeout as a backup fault response. The TB6612FNG driver receives two direction lines plus a PWM/enable line and drives the motor bidirectionally. Protection and support parts include local decoupling capacitors on the MCU and motor driver, bulk capacitance on the 12 V rail near the driver, a fuse on the 12 V input, and a shared ground reference between logic and motor power.

Main components

- U1: ATtiny1616 8-bit microcontroller for keypad scanning and state logic
- U2: TB6612FNG dual H-bridge used as one DC motor channel
- U3: LM2596 buck converter module adjusted to 5 V
- K1: 4x4 matrix keypad
- M1: 12 V geared DC motor coupled to latch mechanism
- S1: Locked-position limit switch
- S2: Unlocked-position limit switch
- F1: 2 A input fuse
- C1/C2/C3/C4: bulk and local decoupling capacitors

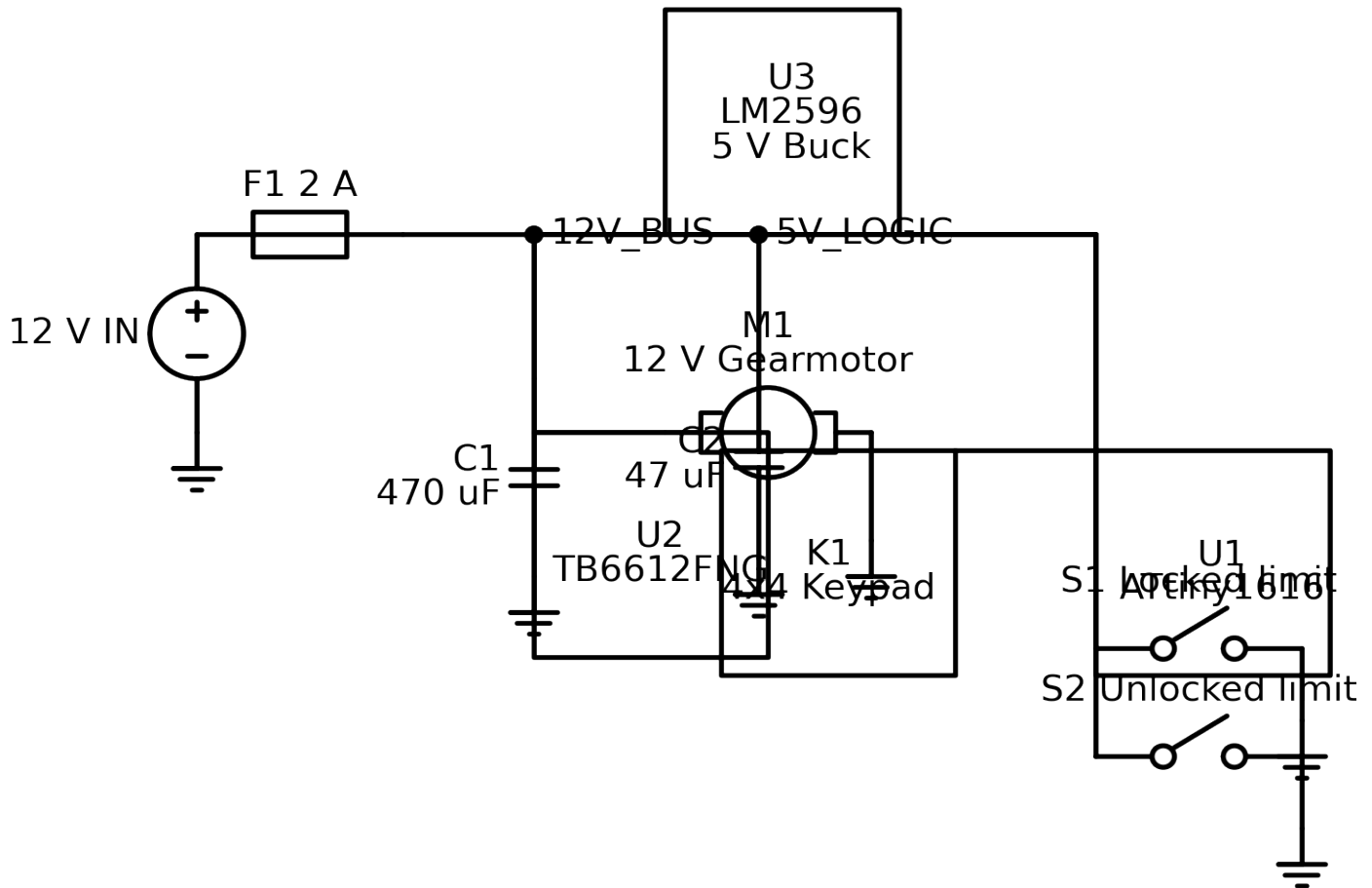
Firmware need

Yes. Firmware is required to scan the keypad matrix, compare entered PIN codes, command lock/unlock motor direction, monitor both limit switches, enforce a motor timeout, optionally debounce keys and switches, and optionally drive a status LED or buzzer.

Important unknowns / verification needed

- Real motor running and stall current
- Mechanical torque needed to move the latch
- Exact keypad model and connector style
- Whether fail-secure or fail-safe behavior is wanted on power loss
- Whether tamper sensors, battery backup, or door-position sensing are required
- Whether a higher-security product should use a certified safe lock assembly instead of a custom mechanism

Schematic



Bill of Materials

Refs	Qty	Description	Part number	Required specs	Package	Unit price	Ext. price	Supplier	URL
U1	1	Microcontroller	ATTiny1616-SFR	8-bit AVR MCU, 16 KB Flash, SOIC-20	SOT-23	\$0.95	\$0.95	Microchip / DigiKey	https://www.microchip.com
U2	1	Motor driver	TB6612FNG	Dual H-bridge, 4.5-13.5 V motor supply	SOP-24	\$2.50	\$2.50	Toshiba / breakout module / solderless	https://www.semiconductors.com
U3	1	Buck converter module	LM2596 module	12 V in to regulated 5 V out, >=0.5 A load	Module	\$1.50	\$1.50	Generic module	https://www.ti.com/lit/ds/s
K1	1	User input keypad	4x4 matrix keypad	Momentary matrix keypad, 8-wire interface	Panel/membrane	\$3.00	\$3.00	Generic	https://www.kiwi-electronics.com
M1	1	Actuator motor	12 V geared DC motor	Low-speed gearmotor sized to latch to wire-matrix	Wire-matrix	\$12.00	\$12.00	Generic	https://www.pololu.com/catalog
S1,S2	2	End-of-travel switches	KW12-3A-style microswitch	SPDT or SPST lever microswitch, use Redomach	Redomach	\$0.50	\$1.00	Generic	https://www.kel-switch.com
F1	1	Input protection fuse	2 A blade or inline fuse	Protect 12 V input wiring against shorts	Blade/fritter	\$1.00	\$1.00	Generic	https://www.littelfuse.com
C1	1	Bulk input capacitor	470 uF 25 V electrolytic	12 V rail bulk decoupling near H-bridge	Radial	\$0.40	\$0.40	Generic	https://www.digikey.com/
C2	1	Logic bulk capacitor	47 uF 10 V electrolytic	5 V rail bulk decoupling near MCU	Radial	\$0.20	\$0.20	Generic	https://www.digikey.com/
C3,C4	2	Decoupling capacitors	0.1 uF ceramic	One each at U1 and U2 supply pins	0805/THT	\$0.05	\$0.10	Generic	https://www.digikey.com/
R1,R2	2	Pull-up resistors	10 kohm	Optional external pull-ups for limit switches	0805	\$0.02	\$0.04	Generic	https://www.digikey.com/
J1-J4	4	Connectors/terminals	Screw terminals / headers	12 V input, motor, switches, keypad/UART access	PCB	\$2.00	\$2.00	Generic	https://www.digikey.com/

Estimated total BOM cost: about \$24.69 excluding enclosure, PCB, wiring, shipping, taxes, and any higher-torque or higher-security mechanical hardware.

Limitations and disclaimer: this package is a conceptual design aid, not a certified secure-safe lock or production release. Real safe applications need mechanical force analysis, stall-current verification, EMI/ESD review, thermal checks, abuse testing, and threat modeling. Supplier pricing and stock were not comprehensively verified live; links are representative starting points.